

Ankur Desai

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EDUCATION

Michigan State University

B.S. in Computer Science

Expected Graduation, 2027

East Lansing, MI

- **GPA:** 3.5 / 4.0
- **Relevant Coursework:** Linear Algebra, Discrete Mathematics, Object-Oriented Software Design, Computer Organization and Architecture, Data Structures & Algorithms

EXPERIENCE

Claude Builder Club @ MSU

Vice President & Co-Founder

Oct. 2025 – Present

East Lansing, MI

- Eliminated manual onboarding entirely by automating member provisioning (GitHub API + Slack SDK) to grant full repo and workspace access in 30 seconds, cutting setup time 97% across 100+ members.
- Replaced paper sign-in sheets with a QR-driven check-in system backed by AWS S3 token caching, enabling real-time attendance tracking and cutting processing time by 80%.
- Enforced production-grade quality via GitHub Actions CI/CD with 90% coverage gates across 30+ tests, reducing bug escape rate by 65%.

MindMosaic

Technical Lead

Aug. 2025 – Nov. 2025

East Lansing, MI

- Led a 6-engineer team building a full-stack knowledge graph application; owned sprint planning, PR review and merge authority, and final architecture decisions across the entire stack.
- Authored all CI/CD pipelines and Dockerfiles, containerizing the polyglot Spring Boot + Angular + Neo4j stack and enforcing branch protection with automated test gates on every production merge.
- Designed dual-database persistence layer (PostgreSQL + Neo4j) for multi-hop graph traversals; achieved 3× faster performance than SQL self-joins and cut p95 query latency 48% through relationship-type indexing.

PROJECTS

Mira — Agentic Voice Assistant | *Next.js, FastAPI, MongoDB, Redis, Celery*

May 2025 – Nov. 2025

- Built multi-client agentic voice assistant that batches natural dialogue, resolves implicit speaker intent, and dispatches real tasks across concurrent voice streams.
- Fine-tuned a Sentence-BERT intent classification head (SVM, 12-class taxonomy) on 200 synthetic utterances, achieving 91% F1; replaced regex-based task routing, eliminating an entire class of dispatch failures.
- Trained a spaCy NER component on 180 annotated examples to extract TASK, PERSON, TIME, and APP entities, improving recall 34% over the `en_core_web_trf` baseline.
- Reduced speaker diarization error rate from 18.4% to 9.1% via x-vector threshold calibration; lazy-loaded Lambda inference cutting cold-start latency by 86% (6,643ms → 930ms).

WizViz — SpartaHack X | *Python, MediaPipe, OpenCV, TensorFlow, NumPy*

Feb. 2025

- Won Interactive Media Track: gesture-controlled wizard duel where full-body pose estimation drives spell casting, blocking, and ability triggering against an AI opponent.
- Built a full-body pose estimation pipeline using MediaPipe Pose to extract and classify 33 skeletal landmarks into discrete game actions; optimized the inference loop to sustain 60 FPS with sub-20ms input latency.
- Implemented a custom 2D vector physics engine with AABB collision detection to project live 3D skeletal coordinates into game state, maintaining <5px positional drift at 60 FPS.

TECHNICAL SKILLS

Languages Python, Java, JavaScript/TypeScript, SQL, Cypher, Golang, C++, Bash
ML & AI PyTorch, scikit-learn, Sentence-BERT, spaCy, MediaPipe, Whisper
Frameworks FastAPI, Spring Boot, Node.js, React, Next.js, Redis, Celery
Databases & Infrastructure PostgreSQL, Neo4j, MongoDB, AWS (Lambda, S3), GitHub Actions